

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MAJOR PROJECT

STROKESENSE: ML-DRIVEN STROKE RISK PREDICTION

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OBJECTIVE :

The objective of the **College Bus Tracking System** is to develop a real-time transport-information and late-arrival alert platform for college bus services. The system aims to allow students and faculty to track college buses without creating an account, while providing authenticated drivers with GPS-based trip tracking and administrators with complete transport management facilities. It predicts bus arrival times at stops and the college and provides clear ETA information even when live GPS data is unavailable or unreliable. The system also aims to identify delayed routes and notify the appropriate class advisors about students assigned to the affected route. The system is designed to improve transportation visibility, provide early warnings about delays, reduce manual transport management efforts, and support reliable decision-making. It does not perform daily boarding attendance tracking.

INTRODUCTION:

The College Bus Tracking System is a real-time transportation management platform designed to improve the efficiency and visibility of college bus services. Students and faculty often depend on fixed bus schedules, but actual arrival times can change because of traffic, route deviations, GPS problems, or network interruptions.important

The system consists of three main applications: a **React Native passenger application**, a **React Native driver application**, and a **React web-based administrator panel**. The driver application securely sends GPS information, while the passenger application displays the current bus location and ETA. The administrator panel manages routes, stops, schedules, drivers, advisors, passenger rosters, alerts, and system operations.

ABSTRACT:

The **College Bus Tracking System** is a real-time transportation management platform designed to improve the reliability and efficiency of college bus services. It helps students and faculty track buses, view routes, check schedules, and receive estimated arrival times without creating an account. The system consists of a passenger application, authenticated driver application, and web-based administrator portal. Drivers securely share GPS locations, while administrators manage routes, stops, schedules, drivers, advisors, and passenger rosters. The system uses **React Native/Expo**, **React**, **Node.js/Express**, **Socket.IO**, **PostgreSQL**, and **Prisma** for real-time tracking and data management. It calculates ETA using GPS data, route progress, stop coordinates, and scheduled timings. It also handles stale GPS, off-route movement, passed stops, and

unavailable live data with clear status messages. When a bus is predicted to arrive late, the system identifies assigned students and notifies the appropriate class advisors. Overall, the system focuses on live tracking, reliable ETA prediction, privacy, and efficient college transportation management.

HARDWARE REQUIREMENTS:

- **Processor:** Intel Core i3 or above
- **RAM:** Minimum 4 GB
- **Hard Disk:** Minimum 10 GB free storage
- **System Type:** 64-bit computer
- **Display:** 1366 × 768 resolution or higher
- **Input Devices:** Keyboard and Mouse
- **Mobile Device:** Android smartphone for driver GPS tracking
- **GPS:** GPS-enabled Android device
- **Network:** Internet connection

SOFTWARE REQUIREMENTS:

- **Frontend:** React, React Native
- **Mobile Framework:** Expo
- **Programming Language:** TypeScript / JavaScript
- **Backend:** Node.js and Express.js
- **Real-Time Communication:** Socket.IO
- **Database:** PostgreSQL
- **ORM:** Prisma
- **Authentication:** JWT and secure cookies
- **API:** REST API
- **Development Environment:** Visual Studio Code
- **Version Control:** Git/GitHub
- **Browser:** Google Chrome / Microsoft Edge / Mozilla Firefox

EXISTING SYSTEM :

In the existing system, college transportation mainly depends on fixed schedules and manual communication. Students may not

know the actual location of the bus or its expected arrival time when there are traffic delays, route changes, GPS problems, or network interruptions. Transport administrators also need to manually manage routes, drivers, schedules, passenger rosters, and delay information.

Traditional methods provide limited real-time visibility and may make it difficult to quickly identify routes that are likely to miss the college-arrival deadline. Class advisors may also not receive timely information about students assigned to a delayed route. These limitations create a need for a centralized and real-time transport management system

PROPOSED SYSTEM:

The proposed **College Bus Tracking System** provides a centralized platform for real-time transportation monitoring. Students and faculty can select routes without logging in, view schedules and stops, track live bus locations, and check ETA. Drivers use an authenticated mobile application to start trips and transmit GPS data. Administrators manage routes, schedules, drivers, advisors, passenger rosters, alerts, and operations through a web dashboard. **Socket.IO** enables real-time updates, while **PostgreSQL** stores transport data. The ETA mechanism uses GPS data and route progress to estimate arrival times. When a bus is predicted to be late, the system identifies affected students and notifies the appropriate class advisors. Privacy controls protect phone numbers and administrator-only information.

REFERENCES :

GUIDE NAME : Ms. Ramesh

COORDINATOR SIGNATURE